



**TEST 1**  
**SEMESTER II, SESSION 2021/2022**

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| <b>COURSE CODE</b> | <b>:</b> | <b>FSPC 0034</b>                |
| <b>COURSE NAME</b> | <b>:</b> | <b>CHEMISTRY II</b>             |
| <b>PROGRAMME</b>   | <b>:</b> | <b>UTM FOUNDATION</b>           |
| <b>DURATION</b>    | <b>:</b> | <b>1 HOUR AND 15 MINUTES</b>    |
| <b>DATE</b>        | <b>:</b> | <b>6 FEBRUARY 2022 (SUNDAY)</b> |

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**INSTRUCTION TO CANDIDATES:**

1. DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.
2. THE PAPER CONSISTS OF 3 QUESTIONS. ANSWER ALL QUESTIONS.
3. ALL ANSWERS HAVE TO BE WRITTEN IN YOUR OWN PIECE/S OF PAPER USING BLUE-COLOURED PEN.
4. PLEASE DON'T FORGET TO WRITE YOUR NAME AND MATRIC CARD NUMBER ON YOUR ANSWER SHEET.
5. PLEASE WRITE THE QUESTION NUMBER CLEARLY ON THE PAPER WHERE YOU ARE ANSWERING.

**WARNING!**

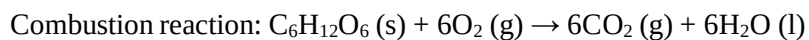
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This booklet consists of **(6)** printed pages including this page.

### QUESTION 1 (10 MARKS)

- a) A student carries out an experiment to determine the enthalpy change of combustion of glucose. In the experiment, 0.460 g of glucose is burned. The energy released is used to heat 100 cm<sup>3</sup> of water from 23.7°C to 41.0°C.
- Calculate the energy released, in kJ during combustion of 0.831 g glucose.  
The specific heat capacity of water = 4.18 Jg<sup>-1</sup>K<sup>-1</sup>. Density of water = 1.00 gcm<sup>-3</sup>.
  - Calculate the amount, in moles, of glucose that is burned. The relative molecular weight of glucose is 180g/mol.
  - Calculate the enthalpy change of combustion of glucose.
  - Explain whether the process is endothermic or exothermic.
- (2 + 1 + 2 + 1 marks)
- b) The standard enthalpy change of combustion can also be determined indirectly. Calculate the standard enthalpy change of combustion of glucose using standard enthalpy changes of formation below.

| Substance                                         | $\Delta_f H$ (kJ mol <sup>-1</sup> ) |
|---------------------------------------------------|--------------------------------------|
| C <sub>6</sub> H <sub>12</sub> O <sub>6</sub> (s) | -1250                                |
| CO <sub>2</sub> (g)                               | -394                                 |
| H <sub>2</sub> O (l)                              | -286                                 |



(4 mark

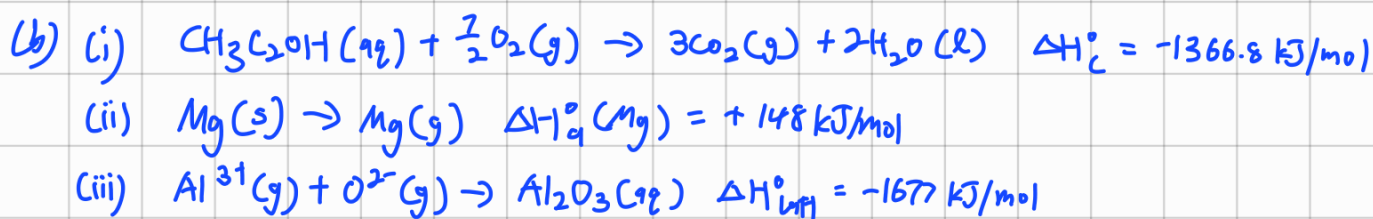
(a) glucose  $C_6H_{12}O_6$ ,  $m = 0.46g$ ,  $V = 100cm^3$ ,  $\Delta T = (41 - 23.7)$ ,  $m.m = 180$   
 mass of water =  $100cm^3 \times \frac{1g}{1cm^3} = 100g$   
 $q = ms\Delta T$   $0.46g \rightarrow 7.231 kJ$   $mol = \frac{0.831}{180}$   
 $= 100(4.18)(17.3)$   $0.831g \rightarrow x kJ$   $= 4.62 \times 10^{-3} mol$   
 $= 7.231 kJ$   $x = 13.06 kJ$

$$\Delta H_c^\circ = \frac{-13.06 kJ}{4.62 \times 10^{-3}} = -2828.9 kJ/mol$$

The process is exothermic because the negative sign and heat energy is released from system to surrounding

(b)  $\Delta H^\circ = \Delta H_{product} - \Delta H_{reactant}$   
 $= (\Delta H_f^\circ H_2O + \Delta H_f^\circ H_2O) - \Delta H_f^\circ C_6H_{12}O_6$   
 $= [6(-394) + 6(-286)] - (-1250)$   
 $= -2830 kJ/mol$

- 2) (a) - particles collide in correct orientation  
 - particles achieve activation Energy ( $E_a$ )  
 - have right geometry  
 - experience effective collision  
 - have right amount of energy



(c) (i) (a) reactants (ii)  $\Delta H^\circ = E_a(\text{forward}) - E_a(\text{reversed})$   
 (b) activation energy  $= 25 - 200$   
 (c) enthalpy change  $= -175 kJ/mol$   
 (d) time

3) (a) (i)  $k[\text{HBr}]^2$ , second order  $\frac{1}{[A_t]} - \frac{1}{[A_0]} = kt$ ,  
 $R = 3.5 \times 10^{-4} = k(1.2)^2$   
 $k = 2.431 \times 10^{-4} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$

(ii)  $n = 2.431 \times 10^{-4} \times 6.02 \times 10^{23}$   
 $= 1.463 \times 10^{20} \text{ hydrogen molecule}$

(b)  $\frac{\text{Rate 2}}{\text{Rate 1}} = \frac{2}{1} = \frac{0.2^x}{0.1^x}$ ,  $2 = 2^x$ ,  $x = 1$

$\frac{\text{Rate 3}}{\text{Rate 1}} = \frac{4}{1} = \frac{0.2^y}{0.1^y}$ ,  $4 = 2^y$ ,  $y = 2$

$\frac{\text{Rate 4}}{\text{Rate 1}} = \frac{1}{1} = \frac{0.2^z}{0.1^z}$ ,  $1 = 2^z$ ,  $z = 0$

$R = k[X]^x[Y]^y[Z]^z$   
 $= k[X][Y]^2[Z]^0$

Total order =  $1 + 2 + 0$   
 $= 3$

(c)  $E_a = 75 \text{ kJ/mol}$ ,  $k = 8.8 \text{ s}^{-1}$

$\ln\left(\frac{k_2}{8.8}\right) = \frac{75}{8.314} \left(\frac{37-27}{37 \times 27}\right)$

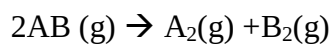
$\ln\left(\frac{k_2}{8.8}\right) = 0.403$

$\frac{k_2}{8.8} = e^{0.403}$

$k_2 = 21.71 \text{ s}^{-1}$

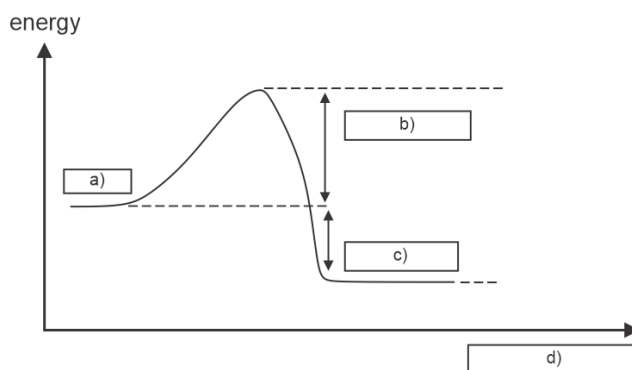
## QUESTION 2 (10 MARKS)

- a) Explain **two** requirements for the effective collision in a chemical reaction. (2 marks)
- b) Write the thermochemical equation for the following standard enthalpy processes:
- Standard enthalpy of combustion of ethanol,  $\Delta H^\circ_c (\text{CH}_3\text{C}_2\text{OH}) = -1366.8$  kJ/mol. Given the reaction releases water and carbon dioxide.
  - Standard atomization energy of magnesium,  $\Delta H^\circ_a (\text{Mg}) = +148$  kJ/mol
  - Standard lattice enthalpy of aluminium oxide,  $\Delta H^\circ_{\text{Latt}} (\text{Al}_2\text{O}_3) = -1677$  kJ/mol
- (1+ 1 + 1 marks)
- c) For the reaction,



Given that the activation energy for the forward and reverse reactions is 25 kJ/mol and 200 kJ/mol, respectively. Below is the energy profile diagram for the reaction.

- Label the energy profile diagram.
- Calculate the enthalpy change for the reaction



(4+1 marks)

**QUESTION 3 (10 MARKS)**

The rate equation for the thermal decomposition of hydrogen bromide is  $\text{rate} = k[\text{HBr}]^2$ . At 250 °C and a concentration of  $1.20 \text{ mol dm}^{-3}$ , the rate of decomposition of hydrogen bromide is  $3.5 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ .

- Calculate the rate constant for the reaction at 250 °C (state its unit).
- Calculate the number of hydrogen molecules formed per second (rate of formation) in  $1 \text{ dm}^3$  of solution. (Avogadro number:  $6.02 \times 10^{23}$ )

(2 + 1 marks)

b) The data below was obtained from the reaction  $\text{X} + \text{Y} + \text{Z} \rightarrow \text{products}$ .

| Experiment | Concentration ( $\text{mol dm}^{-3}$ ) |     |     | Initial rate<br>( $\text{mol dm}^{-3} \text{ s}^{-1}$ ) |
|------------|----------------------------------------|-----|-----|---------------------------------------------------------|
|            | X                                      | Y   | Z   |                                                         |
| 1          | 0.1                                    | 0.1 | 0.1 | 1.0                                                     |
| 2          | 0.2                                    | 0.1 | 0.1 | 2.0                                                     |
| 3          | 0.1                                    | 0.2 | 0.1 | 4.0                                                     |
| 4          | 0.1                                    | 0.1 | 0.2 | 1.0                                                     |

- Find the order of reaction with respect to X, Y and Z.
- Write the rate equation (rate law) and determine the total order of reaction

(3 + 2 marks)

c) The activation energy of a certain first order reaction is  $75 \text{ kJ mol}^{-1}$ . At 27 °C, the rate constant of the reaction is  $8.8 \text{ s}^{-1}$ . What is the rate constant of the reaction if the temperature is increased to 37 °C? ( $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$ )

(2 marks)

|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| 18    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 2    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 18    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |        |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| VIIIA |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | He   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Ne    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Ar    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Kr     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Xe     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Rn    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 4.00 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 20.18 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 39.95 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 83.80  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 131.29 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | (222) |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 10   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 17    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 35.45 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 79.90  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 126.9  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | (210) |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | IVA  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 6     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 28.09 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 72.61  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 118.71 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 82    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 207.2 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 13   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 14    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 14    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 49     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 50     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 81    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 204.4 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | IIIA |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 5     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 10.81 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 69.72  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 114.82 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 81    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 204.4 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 13   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Al    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 13    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 31     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 49     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 81    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 204.4 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 12   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | IB    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 30    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 65.39  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 112.41 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 80    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 200.6 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 10   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 10    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 28    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 58.69  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 106.42 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 78    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 195.1 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 9    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | VIIIB |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 27    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 58.93  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 102.91 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 77    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 192.2 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 8    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 8     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 26    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 55.85  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 101.07 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 76    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 190.2 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 7    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | VIIIB |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 25    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 54.94  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | (98)   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 75    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 186.2 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 6    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | VIB   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 24    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 52.00  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 95.94  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 74    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 183.9 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 5    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | VB    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 23    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 50.94  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 92.91  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 73    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 180.9 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 4    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | IVB   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 22    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 47.88  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 91.22  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 72    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 178.5 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 3    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | IIIB  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 21    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 44.96  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 88.91  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 57    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 138.9 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 20   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Ca    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 20    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 40.08  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 87.62  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 56    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 137.3 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 19   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | K     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 19    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 39.1   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 85.47  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 55    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 132.9 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 37   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Rb    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 37    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 85.47  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 85.47  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 85.47 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 85.47 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 38   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Sr    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 38    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 87.62  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 87.62  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 87.62 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 87.62 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 39   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | K     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 39    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 39.1   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 39.1   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 39.1  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 39.1  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 40   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Ca    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 40    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 40.08  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 40.08  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 40.08 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 40.08 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 41   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Mg    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 12    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 24.31  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 24.31  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 24.31 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 24.31 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 11   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Na    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 11    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 22.99  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 22.99  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 22.99 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 22.99 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 3    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Li    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 3     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 6.94   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 6.94   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 6.94  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 6.94  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 4    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Be    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 4     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 9.01   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 9.01   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 9.01  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 9.01  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | H     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1.01   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1.01   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1.01  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1.01  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | IA    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

|                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| 58                 | 59                 | 60                 | 61                 | 62                 | 63                 | 64                 | 65                 | 66                 | 67                 | 68                 | 69                 | 70                 | 71                 |
| <b>Ce</b><br>140.1 | <b>Pr</b><br>140.9 | <b>Nd</b><br>144.2 | <b>Pm</b><br>(145) | <b>Sm</b><br>150.4 | <b>Eu</b><br>152.0 | <b>Gd</b><br>157.3 | <b>Tb</b><br>158.9 | <b>Dy</b><br>162.5 | <b>Ho</b><br>164.9 | <b>Er</b><br>167.3 | <b>Tm</b><br>168.9 | <b>Yb</b><br>173.0 | <b>Lu</b><br>175.0 |
| 90                 | 91                 | 92                 | 93                 | 94                 | 95                 | 96                 | 97                 | 98                 | 99                 | 100                | 101                | 102                | 103                |
| <b>Th</b><br>232.0 | <b>Pa</b><br>(231) | <b>U</b><br>238.0  | <b>Np</b><br>(237) | <b>Pu</b><br>(244) | <b>Am</b><br>(243) | <b>Cm</b><br>(247) | <b>Bk</b><br>(247) | <b>Cf</b><br>(251) | <b>Es</b><br>(252) | <b>Fm</b><br>(257) | <b>Md</b><br>(258) | <b>No</b><br>(259) | <b>Lr</b><br>(260) |

## LIST OF SELECTED CONSTANT VALUES

|                            |                    |   |                                                                                                |
|----------------------------|--------------------|---|------------------------------------------------------------------------------------------------|
| Speed of light in a vacuum | $c$                | = | $3.0 \times 10^8 \text{ m s}^{-1}$                                                             |
| Avogadro's number          | $N_A$              | = | $6.02 \times 10^{23} \text{ mol}^{-1}$                                                         |
| Faraday constant           | $F$                | = | $9.65 \times 10^4 \text{ C mol}^{-1}$                                                          |
| Planck constant            | $h$                | = | $6.6256 \times 10^{-34} \text{ J s}$                                                           |
| Reduced Planck constant    | $\hbar$            | = | $1.054 \times 10^{-34} \text{ J s}$                                                            |
| Rydberg constant           | $R_H$              | = | $1.097 \times 10^7 \text{ m}^{-1}$<br>$= 2.18 \times 10^{-14} \text{ J}$                       |
| Molar of gases constant    | $R$                | = | $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$<br>$= 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$ |
| Boltzmann constant         | $k$                | = | $1.3807 \times 10^{-27} \text{ J K}^{-1}$                                                      |
| Mass of proton             | $M_p$              | = | $1.672 \times 10^{-27} \text{ kg}$                                                             |
| Electronic Bohr magneton   | $m_B$              | = | $9.2741 \times 10^{-24} \text{ J T}^{-1}$                                                      |
| Nuclear Bohr magneton      | $b_N$              | = | $5.05 \times 10^{-27} \text{ J T}^{-1}$                                                        |
| Vapour pressure of water   | $P_{\text{water}}$ | = | 23.8 torr                                                                                      |
| Electron charge            | $e^-$              | = | $1.602 \times 10^{-19} \text{ C}$                                                              |

## UNIT AND CONVERSION FACTOR

|          |                                                                                                                                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Energy   | $1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$<br>$1 \text{ calorie} = 4.184 \text{ Joule}$<br>$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$<br>$1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$ |
| Pressure | $1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$                                                                                                                                   |